19.2 Solution of Equations by Iteration

For each of the remaining sections of this chapter, we select basic kinds of problems and discuss numeric methods on how to solve them. The reader will learn about a variety of important problems and become familiar with ways of thinking in numerical analysis.

Perhaps the easiest conceptual problem is to find solutions of a single equation

$$f(x) = 0, \tag{1}$$

where f is a given function. A **solution** of (1) is a number $x = s$ such that $f(s) = 0$. Here, s suggests "solution," but we shall also use other letters.

It is interesting to note that the task of solving (1) is a question made for numeric algorithms, as in general there are no direct formulas, except in a few simple cases.

Examples of single equations are $x^3 + x = 1$, $\sin x = 0.5x$, $\tan x = x$, $\cosh x = \sec x$, $\cosh x \cos x = -1$, which can all be written in the form of (1). The first of the five equations is an **algebraic equation** because the corresponding f is a polynomial. In this case the solutions are called **roots** of the equation and the solution process is called *finding roots.* The other equations are **transcendental equations** because they involve transcendental functions.

There are a very large number of applications in engineering, where we have to solve a single equation (1). You have seen such applications when solving characteristic equations in Chaps. 2, 4, and 8; partial fractions in Chap. 6; residue integration in Chap. 16, finding eigenvalues in Chap. 12, and finding zeros of Bessel functions, also in Chap. 12. Moreover, methods of finding roots are very important in areas outside of classical engineering. For example, in finance, the problem of determining how much a bond is worth amounts to solving an algebraic equation.

To solve (1) when there is no formula for the exact solution available, we can use an approximation method, such as an **iteration method**. This is a method in which we start from an initial guess x_0 (which may be poor) and compute step by step (in general better and better) approximations $x_1, x_2, \cdots$ of an unknown solution of (1). We discuss three such methods that are of particular practical importance and mention two others in the problem set.

It is very important that the reader understand these methods and their underlying ideas. The reader will then be able to select judiciously the appropriate software from among different software packages that employ variations of such methods and not just treat the software programs as "black boxes."

In general, iteration methods are easy to program because the computational operations are the same in each step—just the data change from step to step—and, more importantly, if in a concrete case a method converges, it is stable in general (see Sec. 19.1).

Fixed-Point Iteration for Solving Equations $f(x) = 0$

Note: Our present use of the word "fixed point" has absolutely nothing to do with that in the last section.

By some *algebraic steps* we transform (1) into the form

$$x = g(x). \tag{2}$$

Then we choose an x_0 and compute $x_1 = g(x_0)$, $x_2 = g(x_1)$, and in general

$$x_{n+1} = g(x_n) \qquad (n = 0, 1, \cdots). \tag{3}$$

A solution of (2) is called a **fixed point** of g, motivating the name of the method. This is a solution of (1), since from $x = g(x)$ we can return to the original form $f(x) = 0$. From (1) we may get several different forms of (2). The behavior of corresponding iterative sequences $x_0, x_1, \cdots$ may differ, in particular, with respect to their speed of convergence. Indeed, some of them may not converge at all. Let us illustrate these facts with a simple example.

EXAMPLE 1 An Iteration Process (Fixed-Point Iteration)

Set up an iteration process for the equation $f(x) = x^2 - 3x + 1 = 0$. Since we know the solutions

$$x = 1.5 \pm \sqrt{1.25}, \quad \text{thus} \quad 2.618034 \quad \text{and} \quad 0.381966,$$

we can watch the behavior of the error as the iteration proceeds.

Solution. The equation may be written

$$\text{(4a)} \qquad x = g_1(x) = \tfrac{1}{3}(x^2 + 1), \qquad \text{thus} \qquad x_{n+1} = \tfrac{1}{3}(x_n^2 + 1).$$

If we choose $x_0 = 1$, we obtain the sequence (Fig. 426a; computed with 6S and then rounded)

$$x_0 = 1.000, \quad x_1 = 0.667, \quad x_2 = 0.481, \quad x_3 = 0.411, \quad x_4 = 0.390, \cdots$$

which seems to approach the smaller solution. If we choose $x_0 = 2$, the situation is similar. If we choose $x_0 = 3$, we obtain the sequence (Fig. 426a, upper part)

$$x_0 = 3.000, \quad x_1 = 3.333, \quad x_2 = 4.037, \quad x_3 = 5.766, \quad x_4 = 11.415, \cdots$$

which diverges.

Our equation may also be written (divide by x)

$$\text{(4b)} \qquad x = g_2(x) = 3 - \frac{1}{x}, \qquad \text{thus} \qquad x_{n+1} = 3 - \frac{1}{x_n},$$

and if we choose $x_0 = 1$, we obtain the sequence (Fig. 426b)

$$x_0 = 1.000, \quad x_1 = 2.000, \quad x_2 = 2.500, \quad x_3 = 2.600, \quad x_4 = 2.615, \cdots$$

which seems to approach the larger solution. Similarly, if we choose $x_0 = 3$, we obtain the sequence (Fig. 426b)

$$x_0 = 3.000, \quad x_1 = 2.667, \quad x_2 = 2.625, \quad x_3 = 2.619, \quad x_4 = 2.618, \cdots.$$

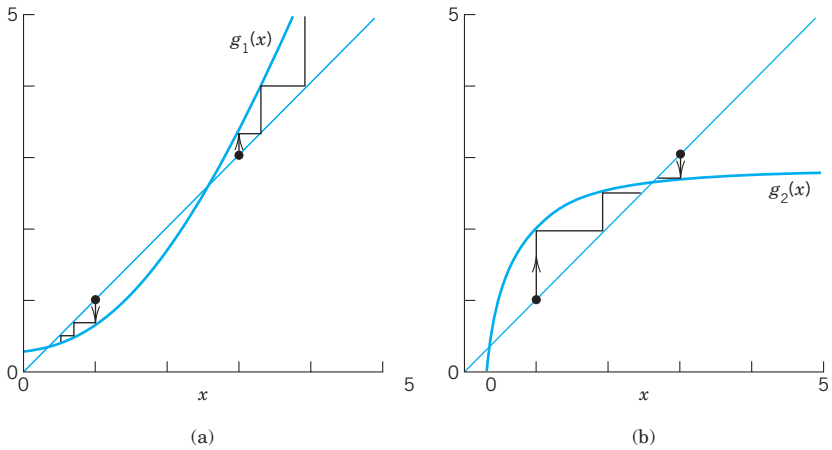


Fig. 426. Example 1, iterations (4a) and (4b)

Our figures show the following. In the lower part of Fig. 426a the slope of $g_1(x)$ is less than the slope of $y = x$, which is 1, thus $|g_1'(x)| < 1$, and we seem to have convergence. In the upper part, $g_1(x)$ is steeper ($g_1'(x) > 1$) and we have divergence. In Fig. 426b the slope of $g_2(x)$ is less near the intersection point ($x = 2.618$, fixed point of g_2, solution of $f(x) = 0$), and both sequences seem to converge. From all this we conclude that convergence seems to depend on the fact that, in a neighborhood of a solution, the curve of $g(x)$ is less steep than the straight line $y = x$, and we shall now see that this condition $|g'(x)| < 1$ (= slope of $y = x$) is sufficient for convergence. ■

An iteration process defined by (3) is called **convergent** for an x_0 if the corresponding sequence $x_0, x_1, \cdots$ is convergent.

A sufficient condition for convergence is given in the following theorem, which has various practical applications.

THEOREM 1

Convergence of Fixed-Point Iteration

Let $x = s$ be a solution of $x = g(x)$ and suppose that g has a continuous derivative in some interval J containing s. Then, if $|g'(x)| \leqq K < 1$ in J, the iteration process defined by (3) *converges for any x_0 in J. The limit of the sequence $\{x_n\}$ is s.*

PROOF By the mean value theorem of differential calculus there is a t between x and s such that

$$g(x) - g(s) = g'(t)(x - s) \qquad (x \text{ in } J).$$

Since $g(s) = s$ and $x_1 = g(x_0), x_2 = g(x_1), \cdots$, we obtain from this and the condition on $|g'(x)|$ in the theorem

$$|x_n - s| = |g(x_{n-1}) - g(s)| = |g'(t)||x_{n-1} - s| \leqq K|x_{n-1} - s|.$$

Applying this inequality n times, for $n, n - 1, \cdots, 1$ gives

$$|x_n - s| \leqq K|x_{n-1} - s| \leqq K^2|x_{n-2} - s| \leqq \cdots \leqq K^n|x_0 - s|.$$

Since $K < 1$, we have $K^n \to 0$; hence $|x_n - s| \to 0$ as $n \to \infty$. ■

We mention that a function g satisfying the condition in Theorem 1 is called a **contraction** because $|g(x) - g(v)| \leqq K|x - v|$, where $K < 1$. Furthermore, K gives information on the speed of convergence. For instance, if $K = 0.5$, then the accuracy increases by at least 2 digits in only 7 steps because $0.5^7 < 0.01$.

EXAMPLE 2 **An Iteration Process. Illustration of Theorem 1**

Find a solution of $f(x) = x^3 + x - 1 = 0$ by iteration.

Solution. A sketch shows that a solution lies near $x = 1$. **(a)** We may write the equation as $(x^2 + 1)x = 1$ or

$$x = g_1(x) = \frac{1}{1 + x^2}, \qquad \text{so that} \qquad x_{n+1} = \frac{1}{1 + x_n^2}. \qquad \text{Also} \qquad |g_1'(x)| = \frac{2|x|}{(1 + x^2)^2} < 1$$

for any x because $4x^2/(1 + x^2)^4 = 4x^2/(1 + 4x^2 + \cdots) < 1$, so that by Theorem 1 we have convergence for any x_0. Choosing $x_0 = 1$, we obtain (Fig. 427)

$$x_1 = 0.500, \quad x_2 = 0.800, \quad x_3 = 0.610, \quad x_4 = 0.729, \quad x_5 = 0.653, \quad x_6 = 0.701, \cdots.$$

The solution exact to 6D is $s = 0.682328$.

(b) The given equation may also be written

$$x = g_2(x) = 1 - x^3. \qquad \text{Then} \qquad |g_2'(x)| = 3x^2$$

and this is greater than 1 near the solution, so that we cannot apply Theorem 1 and assert convergence. Try $x_0 = 1, x_0 = 0.5, x_0 = 2$ and see what happens.

The example shows that the transformation of a given $f(x) = 0$ into the form $x = g(x)$ with g satisfying $|g'(x)| \leqq K < 1$ may need some experimentation. ■

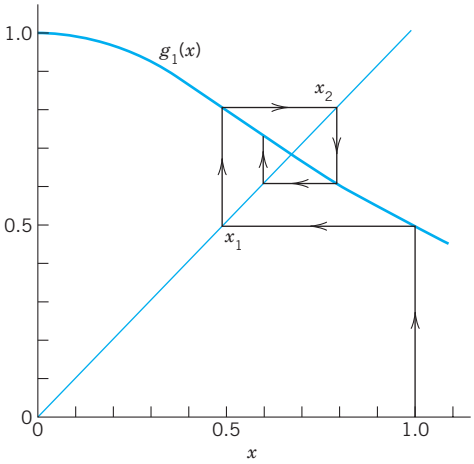


Fig. 427. Iteration in Example 2

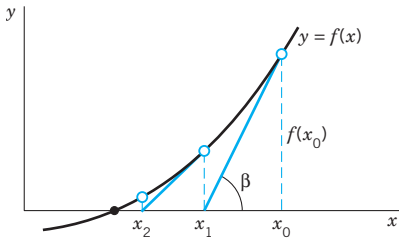


Fig. 428. Newton's method

Newton's Method for Solving Equations $f(x) = 0$

Newton's method, also known as **Newton–Raphson's method**,[1] is another iteration method for solving equations $f(x) = 0$, where f is assumed to have a continuous derivative f'. The method is commonly used because of its simplicity and great speed.

The underlying idea is that we approximate the graph of f by suitable tangents. Using an approximate value x_0 obtained from the graph of f, we let x_1 be the point of intersection of the x-axis and the tangent to the curve of f at x_0 (see Fig. 428). Then

$$\tan \beta = f'(x_0) = \frac{f(x_0)}{x_0 - x_1}, \qquad \text{hence} \qquad x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}.$$

In the second step we compute $x_2 = x_1 - f(x_1)/f'(x_1)$, in the third step x_3 from x_2 again by the same formula, and so on. We thus have the algorithm shown in Table 19.1. Formula (5) in this algorithm can also be obtained if we algebraically solve Taylor's formula

$$(5^*) \qquad f(x_{n+1}) \approx f(x_n) + (x_{n+1} - x_n) f'(x_n) = 0.$$

[1] JOSEPH RAPHSON (1648–1715), English mathematician who published a method similar to Newton's method. For historical details, see Ref. [GenRef2], p. 203, listed in App. 1.

Table 19.1 Newton's Method for Solving Equations $f(x) = 0$

ALGORITHM NEWTON $(f, f', x_0, \epsilon, N)$

This algorithm computes a solution of $f(x) = 0$ given an initial approximation x_0 (starting value of the iteration). Here the function $f(x)$ is continuous and has a continuous derivative $f'(x)$.

INPUT: f, f', initial approximation x_0, tolerance $\epsilon > 0$, maximum number of iterations N.

OUTPUT: Approximate solution x_n $(n \leqq N)$ or message of failure.

For $n = 0, 1, 2, \cdots, N - 1$ do:

1 Compute $f'(x_n)$.

2 If $f'(x_n) = 0$ then OUTPUT "Failure." Stop.

[*Procedure completed unsuccessfully*]

3 Else compute

(5) $$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

4 If $|x_{n+1} - x_n| \leqq \epsilon |x_{n+1}|$ then OUTPUT x_{n+1}. Stop.

[*Procedure completed successfully*]

End

5 OUTPUT "Failure". Stop.

[*Procedure completed unsuccessfully after N iterations*]

End NEWTON

If it happens that $f'(x_n) = 0$ for some n (see line 2 of the algorithm), then try another starting value x_0. Line 3 is the heart of Newton's method.

The inequality in line 4 is a **termination criterion**. If the sequence of the x_n converges and the criterion holds, we have reached the desired accuracy and stop. Note that this is just a form of the relative error test. It ensures that the result has the desired number of significant digits. If $|x_{n+1}| = 0$, the condition is satisfied if and only if $x_{n+1} = x_n = 0$, otherwise $|x_{n+1} - x_n|$ must be sufficiently small. The factor $|x_{n+1}|$ is needed in the case of zeros of very small (or very large) absolute value because of the high density (or of the scarcity) of machine numbers for those x.

WARNING! The criterion by itself does not imply convergence. *Example.* The harmonic series diverges, although its partial sums $x_n = \sum_{k=1}^{n} 1/k$ satisfy the criterion because $\lim (x_{n+1} - x_n) = \lim (1/(n + 1)) = 0$.

Line 5 gives another termination criterion and is needed because Newton's method may diverge or, due to a poor choice of x_0, may not reach the desired accuracy by a reasonable number of iterations. Then we may try another x_0. If $f(x) = 0$ has more than one solution, different choices of x_0 may produce different solutions. Also, an iterative sequence may sometimes converge to a solution different from the expected one.

EXAMPLE 3 **Square Root**

Set up a Newton iteration for computing the square root x of a given positive number c and apply it to $c = 2$.

Solution. We have $x = \sqrt{c}$, hence $f(x) = x^2 - c = 0, f'(x) = 2x$, and (5) takes the form

$$x_{n+1} = x_n - \frac{x_n^2 - c}{2x_n} = \frac{1}{2}\left(x_n + \frac{c}{x_n}\right).$$

For $c = 2$, choosing $x_0 = 1$, we obtain

$$x_1 = 1.500000, \quad x_2 = 1.416667, \quad x_3 = 1.414216, \quad x_4 = 1.414214, \cdots.$$

x_4 is exact to 6D. ■

EXAMPLE 4 **Iteration for a Transcendental Equation**

Find the positive solution of $2 \sin x = x$.

Solution. Setting $f(x) = x - 2 \sin x$, we have $f'(x) = 1 - 2 \cos x$, and (5) gives

$$x_{n+1} = x_n - \frac{x_n - 2 \sin x_n}{1 - 2 \cos x_n} = \frac{2(\sin x_n - x_n \cos x_n)}{1 - 2 \cos x_n} = \frac{N_n}{D_n}.$$

n	x_n	N_n	D_n	x_{n+1}
0	2.00000	3.48318	1.83229	1.90100
1	1.90100	3.12470	1.64847	1.89552
2	1.89552	3.10500	1.63809	1.89550
3	1.89550	3.10493	1.63806	1.89549

From the graph of f we conclude that the solution is near $x_0 = 2$. We compute: $x_4 = 1.89549$ is exact to 5D since the solution to 6D is 1.895494. ■

EXAMPLE 5 **Newton's Method Applied to an Algebraic Equation**

Apply Newton's method to the equation $f(x) = x^3 + x - 1 = 0$.

Solution. From (5) we have

$$x_{n+1} = x_n - \frac{x_n^3 + x_n - 1}{3x_n^2 + 1} = \frac{2x_n^3 + 1}{3x_n^2 + 1}.$$

Starting from $x_0 = 1$, we obtain

$$x_1 = 0.750000, \quad x_2 = 0.686047, \quad x_3 = 0.682340, \quad x_4 = 0.682328, \cdots$$

where x_4 has the error $-1 \cdot 10^{-6}$. A comparison with Example 2 shows that the present convergence is much more rapid. This may motivate the concept of the *order of an iteration process,* to be discussed next. ■